

SCRUM in Project Management

Roles, Cycle, Interplay, and Ticketing

Lei Bagsan

College of Information Studies
Saint Louis University
Baguio, Philippines
2231737@slu.edu.ph

Joaquin Caparas

College of Information Studies
Saint Louis University
Baguio, Philippines
2235008@slu.edu.ph

Nichole Escaño

College of Information Studies
Saint Louis University
Baguio, Philippines
2230897@slu.edu.ph

Minette Mandac

College of Information Studies
Saint Louis University
Baguio, Philippines
2235117@slu.edu.ph

ABSTRACT

Scrum is a widely used Agile framework in software development because it emphasizes adaptability, collaboration, and iteration. It was first introduced in 1986 by Takeuchi and Nonaka and later formalized in 1995 by Schwaber and Sutherland. Scrum promotes flexibility, transparency, and continuous improvement, unlike the traditional waterfall model that follows a rigid sequence. Its foundation is the Scrum Team, which includes the Product Owner, Scrum Master, and Developers. Each role has a clear responsibility within the Scrum cycle of sprint planning, daily scrums, reviews, and retrospectives. The Product Owner manages the Product Backlog and sets priorities aligned with stakeholder needs. The Scrum Master ensures principles are followed, supports collaboration, and removes impediments. Developers focus on creating increments of work that meet the Definition of Done. Ticketing systems enhance this cycle by tracking tasks, promoting accountability, and improving communication. Scrum enables value-driven outcomes in dynamic environments through structure, trust, and feedback.

CCS CONCEPTS

• Software and its engineering → Software creation and management → Software development process management → Software development methods → Agile software development

KEYWORDS

agile, scrum, product owner, scrum master, scrum team, sprint

ACM Reference format:

Lei Bagsan, Joaquin Caparas, Nichole Escaño and Minette Mandac. 2025. SCRUM in Project Management: Roles, Cycle, Interplay, and Ticketing.

*All authors contributed equally to this discussion.

†Author Footnote to be captured as Author Note

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

© 2025 Copyright held by the owner/author(s).

1 Introduction to Scrum

Scrum is one of the most widely used Agile frameworks in software development because of its adaptability, focus on collaboration, and iterative approach. Scrum was first introduced in a 1986 Harvard Business Review article, “The New New Product Development Game”, by Hirotaka Takeuchi and Ikujiro Nonaka. The name “*Scrum*” was borrowed from the sport of rugby, where the players form a scrum to move the ball forward together [<https://www.scrum.org/resources/what-scrum-module>, <https://asana.com/resources/what-is-scrum>]. The same principle applies to Scrum for software development, where the team unites to move the product toward its goal. Later, in 1995, Ken Schwaber and Jeff Sutherland formalized and expanded this concept by introducing the Agile Manifesto and the Scrum Development Process. [<https://asana.com/resources/what-is-scrum>]. Unlike the traditional waterfall model that follows a linear and rigid sequence of phases, Scrum promotes flexibility by breaking down projects into smaller, manageable sprints. Each sprint allows the team to experiment, inspect their progress, adapt their methods, and make necessary improvements. This iterative nature reflects the three pillars of Scrum, which are transparency, inspection, and adaptation. These principles ensure that the team can continuously evolve its process while focusing on delivering value [<https://www.scrum.org/resources/what-scrum-module>, <https://asana.com/resources/what-is-scrum>].

1.1 Scrum Team

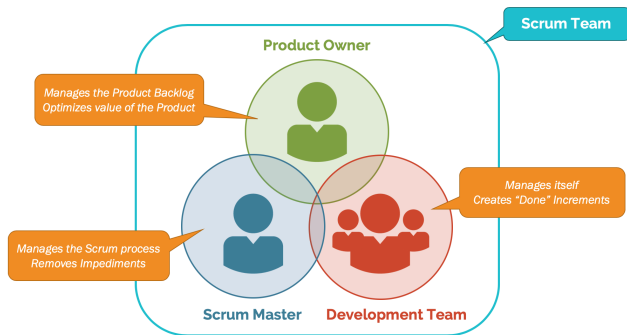


Figure _ : Scrum Team

[<https://www.scrum.org/resources/blog/how-do-3-scrum-roles-promote-self-organization>]

The foundation of Scrum is the Scrum Team. As seen in Figure _ this team comprises three key roles: the Product Owner, the Scrum Master, and the Developers. Each role is essential, and although they have different responsibilities, they are equally important. The Scrum Team is not built on a hierarchy, but rather on collaboration and shared accountability for reaching the Product Goal [<https://www.scrum.org/resources/scrum-team>]. One of the critical characteristics of a Scrum Team is trust. Trust binds the team together and ensures that collaboration flows smoothly. Without trust, the team may experience tension, delays, and unnecessary bottlenecks that hinder productivity. To strengthen trust, members of the Scrum Team are guided by the Scrum Values: Courage, Focus, Commitment, Respect, and Openness [<https://scrumguides.org/scrum-guide.html#scrums-team>]. By working as a cohesive unit, the Scrum Team can align its efforts toward a single objective and maximize the value of the product it delivers. The size of the Scrum Team is intentionally kept small, usually with 10 or fewer members. A team that is too large risks becoming slow, disorganized, and difficult to coordinate, while a team that is too small may not have enough capacity to handle increments of work [<https://scrumguides.org/scrum-guide.html#scrums-team>]. By maintaining balance, the Scrum Team remains flexible enough to adapt and capable of delivering significant outcomes within a Sprint cycle.

1.1.1 The Role of the Product Owner

The Product Owner is the person accountable for ensuring that the team delivers maximum value to the customer and the stakeholders [<https://www.scrum.org/resources/blog/stances-product-owner>]. Product Owners act as a liaison, the bridge between the

team and the outside world, representing the interests and needs of the users. A significant responsibility of the Product Owner is managing the Product Backlog, a dynamic and prioritized list of work items [<https://www.pmi.org/disciplined-agile/product-owner>]. By keeping the backlog clear and well-ordered, the Product Owner ensures that the team always knows what to focus on and why it matters. The Product Owner's role is strategic, as they do not interfere with "how" the team creates it. The success of a Product Owner depends greatly on effective communication. The Product Owner must be available to provide feedback, answer classifications, and explain the reasoning behind priorities [<https://www.mountaingoatsoftware.com/agile/scrum/roles/product-owner>, <https://asana.com/resources/product-owner>]. Therefore, the team avoids confusion and stays aligned with the product's purpose.

1.1.2 The Role of the Scrum Master

The Scrum Master is another critical role within the Scrum Team. Scrum Master is the leader of the Scrum Team. Scrum Masters are concerned about the daily performance of scrum teams. Their primary function is to ensure that Scrum principles, practices, and values are understood and applied correctly. Instead of managing the team directly, the Scrum Master acts as servant-leader who guides and supports the team. The Scrum Masters help the team be self-organized, encourage collaboration, and remove obstacles that may slow down progress [<https://asana.com/resources/scrum-master>]. One of their essential tasks is to educate both the team and the broader organization about Scrum. This is especially valuable when teams are new to Agile methodologies since misunderstandings about concepts like sprints, user stories, or product backlog can easily cause problems. The Scrum Master ensures that the team does not simply follow rules but understand the essence of why Scrum is being implemented. In this way, the Scrum Master contributes not only to the efficiency of the team but also to the long-term growth of the team [<https://asana.com/resources/scrum-master>, <https://www.projectmanager.com/blog/what-is-a-scrum-master-everything-you-need>].

1.1.3 The Role of the Developers

The Developers, meanwhile, are the people who build the actual product. The ideal size for a Scrum Development team is between 3 and 9 people, not including the Scrum Master and Product Owner [<https://resources.scrumalliance.org/Article/scrum-team>, <https://www.simpliaxis.com/resources/scrum-developer-roles-and-responsibilities#heading-0>]. Contrary to the assumption that developers only refer to programmers, this

group is cross-functional and can include designers, testers, business analysts, and other professionals required to create a working product increment for each sprint [https://www.knowledgehut.com/blog/agile/who-is-a-scrum-developer, https://scrumguides.org/scrum-guide.html#developers].

Developers are self-managing, indicating that they decide among themselves how to achieve the Sprint Goal and how to distribute the work. Developers are committed to maintaining quality and following the Definition of Done as their standard of completion. They meet daily to check their progress, adjust their plan if necessary, and hold one another accountable

[https://scrumguides.org/scrum-guide.html#developers, https://resources.scrumalliance.org/Article/scrums-team].

The work of developers is highly collaborative because each member's contribution is essential to the overall product. By combining their diverse skills, they transform the product vision into tangible results that deliver value to the end users [https://resources.scrumalliance.org/Article/scrums-team].

Each of these roles is distinct, the success of a Scrum Team depends on how effectively they interact and collaborate. The Product Owner ensures the team builds the right product by clarifying priorities and goals. The Scrum Master ensures the team works correctly by promoting adherence to Scrum principles and removing barriers. The Scrum Developers ensure that the product is actually built and delivered with quality. If one of these roles fails to perform, the whole framework suffers. For instance, without a clear Product Owner, the team may lose direction, Without a Scrum Master, the team may struggle with process issues and conflicts. Without Developers, no product would exist. Their roles may differ, but are interdependent, and their success is shared.

2 Scrum Cycle

To understand the Scrum lifecycle, it is essential to understand its core element, the sprint [https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf]. A sprint is the fundamental unit of the Scrum lifecycle, defined as a time-boxed event typically lasting one to four weeks, where a team works to create a valuable and usable product increment [https://www.linkedin.com/pulse/role-product-owner-agile-team-lee-eason]. Each sprint is a self-contained iteration with a specific set of activities and artifacts. The Scrum lifecycle is best understood by outlining the key events, commonly referred to as "ceremonies," that occur within and around each sprint.

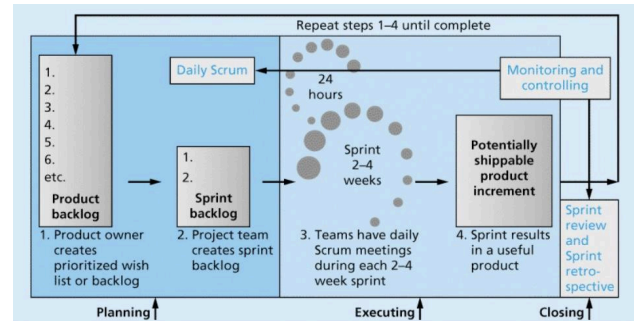


Figure _ : Scrum Lifecycle

The term "ceremony" is used in Scrum to emphasize that these key events are not just ordinary meetings. These are highly structured, purposeful, and time-boxed to ensure maximum value [https://www.projectmanager.com/blog/guide-to-scrum-ceremonies]. While the official Scrum Guide now uses the term "events," "ceremonies" are still widely used in the agile community to highlight their unique nature. The four specific events are discussed below.

2.1 The Product Backlog

Prior to the commencement of a sprint, the Product Backlog must be established [http://dx.doi.org/10.22214/ijraset.2022.44143]. The Product Backlog is handled by the Product Owner, who is responsible for managing and prioritizing it based on business value and stakeholder needs [https://www.linkedin.com/pulse/role-product-owner-agile-team-lee-eason]. The Product Backlog itself is a prioritized, dynamic list of all the features, bug fixes, and tasks required for the product.

2.2 Sprint Planning

This event marks the beginning of a sprint, where the Scrum Team meets to select the highest-priority items from the Product Backlog to work on during the sprint [https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf]. This activity, known as Product Backlog Refinement, is a continuous process essential for a productive sprint session. The Product Owner and the team will add details and estimates to the backlog items. With a well-refined backlog, the team can start planning immediately, which enables the team to initiate planning without delays caused by unclear requirements. With a clear Sprint Goal, the team can work with flexibility and adaptability to adjust their plans while working toward the set goal [http://dx.doi.org/10.22214/ijraset.2022.44143].

2.3 During the Sprint: Daily Scrum

Once the sprint begins, the development team undertakes the selected tasks. Each day, the Scrum Master facilitates a Daily Scrum, a short, 15-minute meeting in which team members share progress updates, identify impediments, and synchronize their activities [<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>]. The Scrum Master is responsible for addressing or removing these impediments, thereby supporting the team's productivity and cohesion.

2.4 Post-Sprint: Review and Retrospective

At the end of each sprint, two events are held: the Sprint Review and the Sprint Retrospective [<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>]. In the Sprint Review, the team presents the completed work to the stakeholders, and the product increment is inspected for feedback. A potentially shippable increment is a critical term used to identify that the completed work during a sprint is in a state ready for release

[<https://www.linkedin.com/pulse/role-product-owner-agile-team-lee-eason>]. It is understood to be fully tested, free of known bugs, and meets the Product Owner's "Definition of Done." The DoD is a clear, shared understanding of what it means for a task to be complete. This marks the commitment that ensures the project's quality and transparency [<https://www.projectmanager.com/blog/guide-to-scrum-ceremonies>].

Following the review, the Scrum Team conducts a Sprint Retrospective to reflect on the sprint itself. The team evaluates what was successful, areas of improvement, and creates a plan to implement these improvements in the next sprint. This process embodies the core agile principle of "inspect and adapt," which is at the heart of Scrum's principle of continuous improvement [<http://dx.doi.org/10.22214/ijraset.2022.44143>].

(ung in-text citation neto is based dun sa references sa 2nd section tab)

3 Roles within the Scrum Cycle

With the Scrum roles and cycle defined, it is understood that the Scrum methodology operates on the integration of the roles within the cycle. Figure _ shows what elements compose Scrum; Roles, Activities, and Artifacts are the main elements of a Scrum. Roles include the Product Owner, the Scrum Master, and the Development Team. The Artifacts include the Product Backlog, Sprint Backlog, and

the Potentially Shippable Product Increment. Lastly, the Activities include the Sprint, Sprint Planning, Daily Scrum, Sprint Execution, Sprint Review, Sprint Retrospective, and Product Backlog Grooming. Scrum being an agile method is inclined to experience challenges in the three areas identified that are critical to be successfully integrated into agile projects; these being (1) development process conflicts, (2) business process conflicts, and (3) people conflicts whereas the people issues are at the heart of the agile transformation [<https://ieeexplore.ieee.org/abstract/document/8812836>]. With these challenges, the three roles all work together to achieve the objective of a project success.

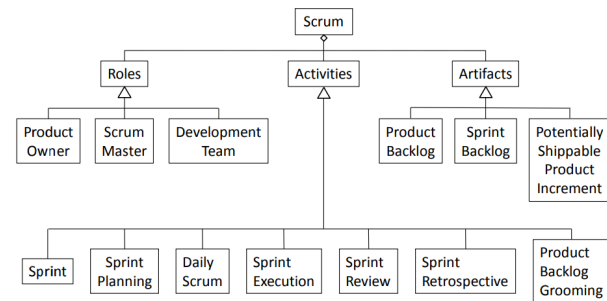


Figure _ : Scrum Composition

3.1 Role and Responsibility of the Product Owner in Scrum Cycle

Product Owners are valued as the link between all stakeholders in a project, and they are the central points of empowered leadership through making decisions on which are to be included in the product backlog, and in what order they should be delivered [<https://www.cs.uic.edu/~i442/Notes/1%20-%20Agile%20and%20Scrum.pdf>]. Being the people that communicate closely with the client—if there is one—they hold a critical position for the project's success. It is perceived that the role of the product owner is the most difficult one to become proficient in [<https://ieeexplore.ieee.org/abstract/document/8812836>], and in addition to that, the role of a Product Owner seems to be neglected in the literature, having most of them dedicated to the role of a Scrum Master [<https://doi.org/10.1002/j.2334-5837.2013.tb03023.x>].

3.1.1 Product Owner During Backlog Preparation and Sprint Planning

Shown in Figure _ are the artifacts wherein the Product Backlog is the primary responsibility of a Product Owner. The Product Backlog is a structured and detailed document that contains a list of requirements that are prioritized by business value. It is the role of the Product Owner to maximize the value by minimizing risk [<https://www.linkedin.com/pulse/art-product-ownership-maxi>]

[mizing-value-minimizing-risk-iacobescu](#)]; to make a product backlog that is manageable and maintainable. An effective Product Owner must produce a product backlog that will result in a clear path of where the Product is heading towards, and it must enable the Scrum Master and the Development Team to a better understanding of what to deliver. To be an effective Product Owner, there are various suggestions given by project management professionals [<https://www.scrum.org/resources/blog/10-tips-product-owners-product-backlog-management>; <https://nevolearn.com/blog/product-owner-role-in-sprint-planning>]; (1) the product backlog is a living artifact that is ever-changing, it is not a definite list; (2) it is important to seek input and feedback from clients, designers, and the development team to optimize the workload and the product delivery; (3) keep in mind the responsibility to maximize value for the customers/clients and users.

3.1.2 Product Owner During Sprint Execution

In the Sprint Execution phase of Scrum, the Product Owner is not required to attend Daily Scrum meetings [<https://www.sustainedagility.com/blog/should-the-product-owner-attend-the-daily-scrum-a-closer-look>]. However, considering the practices in Agile methodology being rooted in collaboration and communication, where also one of the core values in the Agile Manifesto emphasize on individuals and interactions over processes and tools [<https://www.pmi.org/disciplined-agile/agile/theagilemanifesto>]; it can be derived that a Product Owner's participation in this phase allow the Product Owner to gain direct insights on the team's development. This enables the Product Owner to adjust product backlogs, if necessary, to ensure that the team can deliver the most valuable product. It is essential to note that a Product Owner's participation during this phase does not necessarily mean to have a direct involvement, but to observe silently, unless asked for contributions.

3.1.3 Product Owner During Sprint Closing

During the Sprint Closing phase where Review and Retrospective operations are done, the Product Owner has to be present in these meetings. The Product Owner can also invite key stakeholders to gain insights. During Spring Review, it presents an opportunity for the Product Owner to provide feedback whether the delivered features meet the definition of 'done', and to approve if the product is for release to client and users [<https://www.informingscience.org/Publications/4549>; <https://dl.acm.org/doi/abs/10.1145/3183440.3195066>]. The Product owner uses the feedback they receive to reprioritize the Product Backlog, as they now have concrete data from the clients on what they actually want, not what they say they want

[<https://books.google.com.ph/books?id=qZuCDwAAQBAJ&pg=PA3&ots=An1Qo6Jbvk&dq=scrum%20master%20scrum%20agile%20role%20scrum%20master&lr&pg=PA3#v=onepage&q=scrum%20master%20scrum%20agile%20role%20scrum%20master&f=false>]. During the Sprint Retrospective, the relationship with the Scrum team is also strengthened through accepting feedback and assessments from them [<https://www.scrum.org/forum/scrum-forum/15968/product-owner-be-present-sprint-retrospective>], this identifies on what to improve on that could be implemented in the next Sprint [<https://www.romanpichler.com/blog/product-owner-sprint-retrospective/>].

Product Backlog Grooming is not a formal Scrum event, but this is a continuous responsibility of the Product Owner. On an ongoing basis, the Product Owner refines the backlog items for future Sprints, to ensure they are clear and ready for the Development Team to begin working on. This proactive measure ensures that future Sprints can start smoothly without delays [<https://www.skilluped.com/blog/agile-backlog-grooming-scrum-product-owner-2025>].

3.2 Role and Responsibility of the Scrum Master in Scrum Cycle

The Scrum Master is responsible for facilitating the development process, ensuring that the team uses the full range of appropriate agile principles, practices and rules [<https://arxiv.org/pdf/1712.01177>]. This role is crucial for having responsibilities which includes teaching, coaching, mentoring teams, removing impediments, acting as a protector of the team, and more [<https://www.semanticscholar.org/paper/Scrum-Without-a-Scrum-Master-Ereiz-Music/9c02321c42563ea3a6c37c066626a02c81f91792>]. Their entire job is to help the Team go faster; to make the team produce value twice as quickly [<https://books.google.com.ph/books?id=qZuCDwAAQBAJ&pg=PA3&ots=An1Qo6Jbvk&dq=scrum%20master%20scrum%20agile%20role%20scrum%20master&lr&pg=PA3#v=onepage&q=scrum%20master%20scrum%20agile%20role%20scrum%20master&f=false>]. The Scrum Master is a servant-leader for the Scrum Team [<https://scrumguides.org/scrum-guide.html>]. The responsibilities of a Scrum Master may have similarities with the Product Manager (responsible for the success of a specific product [<https://starkephillip.com/product-manager-vs-project-manager-vs-product-owner-vs-program-manager>]) [<https://ieeexplore.ieee.org/document/6170923>]. Also, the role of a Scrum Master is evolving; this can be adapted to create new roles, be shared by different people, and more. [https://www.researchgate.net/publication/301610726_Adapting_Agile_in_a_Globally_Distributed_Software_Development]; <https://dl.acm.org/doi/10.1109/ESEM.2011.49>. In

summary, there is an emerging theme in the literature, that the original description of what role the Scrum Master holds in this method is being adapted, conflated, and possibly corrupted, to suit the needs of organizations. Therefore, in the following subsections, the role of a Scrum Master within the Scrum lifecycle is discussed.

3.2.1 Scrum Master During Sprint Planning

The Scrum Master, having the expertise on all the elements of Scrum coaches everyone on the Scrum processes. In this phase, the Scrum Master's role is to coach the Product Owner on effective Product Backlog management (but not dictate on what to include in the artifact or which features must take highest priority [<https://books.google.com.ph/books?id=Opw3EAAAQBAJ&pg=PP1&ots=uQ5YVY7QtI&dq=scrum%20master%20scrum%20agile%20role%20scrum%20master&lr&pg=PA30#v=onepage&q=scrum%20master%20scrum%20agile%20role%20scrum%20master&f=false>]) and to help in ensuring that there is transparency for the Scrum Artifacts. During the Sprint Planning, the Scrum Master drives the discussion around the Sprint Goal, and helps in the estimation, tasks decomposition, splitting user stories, if needed [<https://scrummastered.com/blog/scrum-master-sprint-planning>]. They protect the team from over-committing and scope creeping. In Sprint Planning, the Scrum Master is not directly involved with what needs to be done, but is there to ensure that the Scrum framework is being implemented properly.

3.2.2 Scrum Master During Sprint Execution

The Daily Scrum is the core of Sprint Execution, and it is orchestrated and is ensured by the Scrum Master to happen [<https://www.atlassian.com/agile/scrum/scrum-master#:~:text=Summary%3A%20The%20scrum%20master%20serves,team%20to%20improve%20their%20workflow.>]. The Scrum Master teaches the Development Team to keep the Daily Scrum within a 15-minute time-box [<https://www.scrum.org/resources/what-is-a-daily-scrum>]. If the Scrum Master is actively working on items in the Spring Backlog, they participate also as Developers [<https://scrumguides.org/scrum-guide.html#sprint-planning>].

3.2.3 Scrum Master During Sprint Closing

The Scrum Master's responsibilities during the Sprint Review and Retrospective align with their broader role as a facilitator and coach [<https://scrumguides.org/scrum-guide.html>]. During the Sprint Review, they have no authority over decisions about the product; but they help the Development Team to prepare to showcase a good quality and "Done" increment of the product [<https://scrumguides.org/scrum-guide.html>]. Furthermore, the Scrum Master can help the Product Owner by understanding key business metrics presented at the

review, such as market conditions, product usage, and projected release dates. This knowledge enables the Scrum Master to better support the Product Owner in managing expectations and to coach the Development Team on optimizing their work to meet deadlines [<https://archive.org/details/succeedingwithag0000cohn/page/n2/mode/1up>].

During the Sprint Retrospective, the Scrum Master is more involved. The Scrum Master seats with the Product Owner and the Development Team, and figures out what went well, what could have gone better, and what the Team wants to change in how they work to make things better and faster next time they do a Sprint [<https://books.google.com.ph/books?id=Opw3EAAAQBAJ&pg=PP1&ots=uQ5YVY7QtI&dq=scrum%20master%20scrum%20agile%20role%20scrum%20master&lr&pg=PA30#v=onepage&q=scrum%20master%20scrum%20agile%20role%20scrum%20master&f=false>]; the Scrum Master is expected to adopt a neutral stance [<https://www.neatro.io/blog/retrospective-scrum-master>]; but if the Scrum Master sees an issue or potential improvement about the Scrum implementation, they provide opinions as the Scrum "coach" [<https://www.scrum.org/resources/blog/what-role-should-scrum-master-take-sprint-retrospective>]. Ultimately, the Sprint Retrospective is the Scrum Master's time to shine as a true coach. By guiding the team to identify their own areas for growth and encouraging them to commit to concrete changes, the Scrum Master transforms valuable insights into tangible action. This ensures the team learns from each Sprint, consistently refining its methods to deliver a better product more efficiently in the future.

The Scrum Master also supports the Product Owner in Product Backlog Grooming. While the Product Owner is ultimately responsible for the backlog's content, the Scrum Master helps to facilitate the refinement process and teaches the team effective techniques for estimating and clarifying items [<https://www.scrum.org/forum/scrum-forum/28709/product-backlog-refinement-scrum-masters-role>]. They ensure the backlog is consistently groomed and that the team understands the value of this ongoing activity, preventing it from becoming a bottleneck.

3.3 Role and Responsibility of the Development Team in Scrum Cycle

The Development Team is the main component of the overall Scrum process, they are the ones responsible for creating a plan for the Sprint, which is called the "Sprint Backlog", and also creating a "Done," the usable Increment for each Sprint

[<https://www.scrum.org/resources/what-is-a-scrum-developer>]. Unlike traditional hierarchies, this team is self-organizing and cross-functional [<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>]. The team has the autonomy to decide how to best accomplish the work needed to be done, and what comprises the team must possess the necessary skills to deliver a complete product without relying on external parties.

3.3.1 Development Team During Sprint Planning

As a self-organizing team, their participation during the Sprint Planning is beyond simply receiving tasks. They engage with the Product Owner to collaboratively define the Sprint Goal, which is the commitment of their upcoming Sprint. With the given Product Backlog, the Development Team, which are often called Developers (not necessarily a software developer [<https://www.scrum.org/resources/what-is-a-scrum-developer>]), but in the context of Information Technology, they mostly are), select items from the list to include in their upcoming Sprint to commit working on, they then decompose the items into smaller work items of one day or less—how the developers will do this is at their sole discretion [<https://resources.scrumalliance.org/Article/sprint-planning-meeting>].—this forms the Sprint Backlog. The autonomous nature of developers in this phase is crucial; they know their own capabilities and velocity, which results in each developer choosing on which parts of the Increment to work on, this ensures their plan is grounded in reality and feasibility.

3.3.2 Development Team During Sprint Execution

During this phase, the Development Team's dedication to the Sprint Backlog comes to fruition. The Daily Scrum, the heart of this phase, is the 15-minute event dedicated for the Developers. In these daily meetings, an inspection to the progress towards the Sprint Goals occurs, and the Developers collaboratively adjust their plans for the next 24 hours. By having these daily check-ins, the Developers quickly identify and address what they can improve regarding their development, this promotes fast decision-making, and this guarantees that their work remains on track and is aligned with their commitment. The Daily Scrum is not the only time that the Developers are allowed to adjust their plan. They frequently get together throughout the day for discussions to re-plan the rest of the Sprint's work [<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>].

The Daily Scrum is not a status report for the Scrum Master or Product Owner, but a strategic session for the Developers. The team is qualified to choose the techniques

and methods that best suit their needs, as long as their vision targets achieving the Sprint Goal. Their autonomy fosters teamwork and communication, and their flexible approach to planning allows them to manage complexity and continuously make progress toward a “Done” Increment [<https://archive.org/details/succeedingwithag0000cohn/page/n2/mode/1up>].

3.3.3 Development Team During Sprint Closing

The Developers present the “Done” Increments to key stakeholders and progress toward the Product Goal is discussed. The Sprint Review is not limited to a presentation, but is a working session for all the people who attended the Sprint Review to collaborate on what to do next

[<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>]. Following this, they lead the discussion in the Sprint Retrospective, where they inspect how the last Sprint went in terms of individuals, interactions, processes, tools, and their Definition of Done. They identify assumptions that led to issues and explore their origins, discussing what went well, what problems they encountered, and how those problems were or were not solved [<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>].

This inspection allows the team to point out the most impactful changes to improve their effectiveness. They then add these improvements to the Sprint Backlog for the next Sprint

[<https://business.adobe.com/blog/basics/sprint-retrospective>], ensuring that lessons learned translate directly into concrete actions. In addition to these formal Scrum events, the Development Team is a critical participant in Product Backlog Grooming. In collaboration with the Product Owner, they help refine and estimate future backlog items. Their technical expertise is essential for breaking down large items into smaller tasks and providing accurate estimates, which ensures the backlog is ready for upcoming Sprints and that the team has a shared understanding of the future work.

The successful application of Scrum's roles, events, and artifacts relies on effective tracking and management of work. While the Scrum framework provides the blueprint for how teams operate, it is the practical use of tools, often through a system of “ticketing,” that makes the process transparent and actionable.

4 Scrum Ticketing

In working with different Agile methodologies, particularly Scrum, one commonly used approach is the ticketing

system, which allows teams to track and manage tasks in the form of tickets. A ticket typically represents a specific task, user story, support request, or bug. These tickets can be placed in the product backlog or directly into a sprint, enabling the team to visualize and analyze progress throughout the development cycle. This helps ensure everyone remains aligned in the Agile process [https://www.meegle.com/en_us/topics/ticketing-system/ticketing-system-for-agile-workflows].

A key benefit of ticketing is its ability to centralize workflow and promote collaboration. Since tickets are visible to all team members, stakeholders can track progress, add comments, and take action when needed. This transparency fosters accountability and helps teams adapt to changes quickly [<https://www.sciencedirect.com/science/article/pii/S016412122200084X>].

In addition to that, Agile methodologies often deal with uncertainty and require strong communication and structured processes. A study highlights that poor communication and lack of effective processes increase the risk of project failure, especially in uncertain environments [https://www.researchgate.net/publication/263503966_Quantitative_Assessment_with_Using_Ticket_Driven_Development_for_Teaching_Scrum_Framework]. By using a ticketing system, teams can mitigate these risks because tickets provide a structured way to organize, prioritize, and monitor tasks, ensuring a smoother development flow.

4.1 Process of Using Ticket in Scrum

Based on the book about A Practical Guide to the Most Popular Agile Process [<https://libdoc.dpu.ac.th/eBook/113642.pdf>], ticket creation begins when a Product Backlog Item is added to the Product Backlog, which can be in the form of user stories, defects, technical tasks, or research spikes. During backlog refinement, the team works with the Product Owner to clarify each ticket, establish acceptance criteria, and ensure that the item is small, clear, and testable. In Sprint Planning, selected tickets from the Product Backlog are moved into the Sprint Backlog. At this stage, larger product backlog items are broken down into tasks, and each item is given an effort estimate, which is commonly used in story points. These estimates help the team prioritize and forecast how much work can be completed in the sprint.

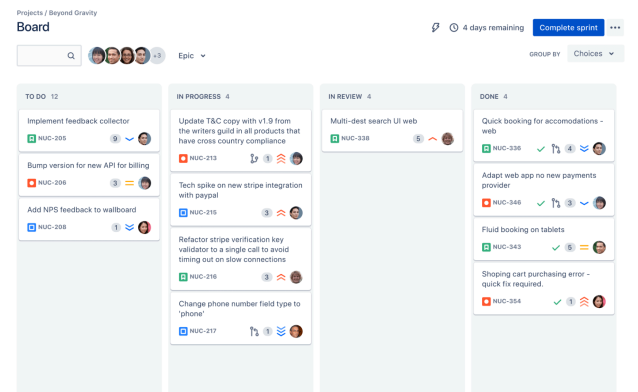


Figure _: Sample Scrum Ticketing System in Jira

<https://wac-cdn.atlassian.com/dam/jcr:3511a84d-0372-43fd-a2af-2005a6eabb3f/Screen-scrum%20board.png?cdnVersion=2945>

Figure _ illustrates how, as a sprint progresses, tickets are actively managed on a Scrum board. Tickets are pulled from the Sprint Backlog and move through stages such as **To Do** → **In Progress** → **In Review** → **Done**, depending on the team's visualization approach. This process provides transparency and enables the team to effectively manage collaboration, track responsibilities, and align their strategy toward achieving the sprint goal. A ticket can only be considered "Done" when it satisfies both its acceptance criteria and the team's Definition of Done, meaning it has been coded, tested, reviewed, and is potentially shippable. At the Sprint Review, completed tickets are demonstrated to stakeholders. Their feedback may result in new tickets, which are added back into the Product Backlog, continuing the cycle of inspection and adaptation [<https://libdoc.dpu.ac.th/eBook/113642.pdf>].

In order to support and carry out these processes, teams use tools such as Jira and Trello to manage tickets and streamline collaboration. Jira is one of the most popular tools, offering rich backlog management, sprint tracking, and reporting features suited for enterprise-level teams [<https://www.atlassian.com/software/jira/templates/sprint-backlog>]. On the other hand, Trello provides a simple, movable, visual board system that is easy to adapt to Scrum, making it ideal for smaller teams or those new to Agile [<https://www.atlassian.com/blog/trello/how-to-scrum-and-trello-for-teams-at-work#:~:text=If%20your%20team%20struggles%20to%20measurable%20progress%20week%20over%20week>].

4.2 Proper Writing of Ticket in Scrum

To have an effective ticket in Agile methodology, particularly Scrum, the team should be able to write and structure their tickets clearly so everyone can easily understand them. Writing tickets hierarchically helps the team break down tasks and understand the requirements more easily. The ticket hierarchy can be structured as an Epic, which is a broad objective or an extensive body of work that outlines a strategic direction and is divided into multiple features. User Story, which is a specific user need derived from an epic, expressed in simple language with clear acceptance criteria. Task or a technical ticket work item from a user story that specifies what must be done to build the feature. By structuring requirements this way in the hierarchy form of tickets, a team can properly assess and understand them, ensuring they are both user-centric and technically actionable
[\[https://www.thinairlabs.ca/post/writing-good-tickets\]](https://www.thinairlabs.ca/post/writing-good-tickets).

Item	Type	Description
As a user, I want the app to be customized to me.	Epic	
As a user, I want to be able to sign up for the system so that I can personalize my experience.	User Story (Story 1: User Signup)	
Build User Database	Technical Ticket	Design and implement the user database schema, including user profiles, secure password storage, and validations. Dependency: Must be completed before building the signup page.
Build Signup Page	Technical Ticket	Develop a responsive signup form with front-end validation and error messaging. Dependency: Uses the user database; design should follow UI/UX best practices.
As a user, I want to be able to log in	User Story (Story 2: User Login)	

to the system so that I can access my personalized features.		
Build Login Page	Technical Ticket	Create the login UI with fields for email/username and password. Dependency: Must align with the signup process design.
Implement Authentication Logic	Technical Ticket	Write backend code to validate user credentials and implement secure session or token-based authentication. Dependency: Requires the user database to be implemented and integrated with the signup process.

Table _ : Sample Hierarchy of Ticket
[\[https://www.thinairlabs.ca/post/writing-good-tickets\]](https://www.thinairlabs.ca/post/writing-good-tickets)

Looking at Table _ from the example of Thin Air Labs' Product Traction team, the way of writing tickets should emphasize the value of what needs to be done. Looking at the epic in Table _, it is a broad objective that can be broken down into user stories. Writing user stories should start by identifying the user or the "who", followed by the "what", which explains the specific feature, and the "why", which is why it is being done. Following this format ensures that the team or anyone reading the ticket will be able to understand and clarify the requirements, avoiding confusion
[\[https://medium.com/@stephan.alex.young/how-i-structure-my-user-story-tickets-4289a5648e06\]](https://medium.com/@stephan.alex.young/how-i-structure-my-user-story-tickets-4289a5648e06) . The user stories will be broken down into smaller tasks, as shown in Table _ . These smaller tasks or technical tickets should be explicitly written, just like in the example, where the user database schema and structure are defined to ensure no task overlap. Technical tickets should explain how, including the technical requirements, validating assumptions, and any dependencies, so that overlapping tasks are avoided and what is needed before proceeding to it. Additionally, they should have defined acceptance criteria to clarify what is needed, and in some cases, begin with a planned research spike
[\[https://www.thinairlabs.ca/post/writing-good-tickets\]](https://www.thinairlabs.ca/post/writing-good-tickets).

Moreover, the team should ensure consistency and standardization when creating tickets by applying best practices and team-defined standards based on the familiarity of the team or the standards of the corporation. This makes tickets easier to interpret, prevents confusion,

saves time, and boosts overall productivity. A prioritization system should also be in place, which includes considering dependencies and assigning story points so the team can identify which tasks must be addressed first. Using ticket management tools such as Jira or Trello helps organize tickets properly while supporting effective monitoring, tracking, and handling of tasks. Lastly, during sprint reviews, it is essential to value feedback and use it to improve future tickets when adding items to the product backlog [<https://www.spinach.ai/content/ticket-management>].

CONCLUSION

However, the effectiveness of Scrum still depends on the team's capability to manage the process and utilize the tools appropriately, since even a well-defined framework requires competent execution to achieve its intended outcomes..

ACKNOWLEDGMENTS

The insights gained from the professors and handlers of the Information Resource Management (IRM) and Software Engineering classes at Saint Louis University (SLU) provided the foundational understanding of Agile and Scrum methodologies that informed this study. In addition to that, the knowledge gained from related literature and available resources played a significant role in shaping the research approach and deepening the researchers' perspective on the topic. Furthermore, the collaboration and contributions of the researchers were essential in completing this work, emphasizing the importance of teamwork in achieving the research purpose.

REFERENCES

- [1] Patricia S. Abril and Robert Plant, 2007. The patent holder's dilemma: Buy, sell, or troll? *Commun. ACM* 50, 1 (Jan, 2007), 36-44. DOI: <https://doi.org/10.1145/1188913.1188915>.
- [2] Sten Andler. 1979. Predicate path expressions. In *Proceedings of the 6th. ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL '79)*. ACM Press, New York, NY, 226-236. DOI:<https://doi.org/10.1145/567752.567774>
- [3] Ian Editor (Ed.). 2007. *The title of book one* (1st. ed.). The name of the series one, Vol. 9. University of Chicago Press, Chicago. DOI:<https://doi.org/10.1007/3-540-09237-4>.
- [4] David Kosiur. 2001. *Understanding Policy-Based Networking* (2nd. ed.). Wiley, New York, NY..